Natural Computing Project Report

Particle Swarm Optimization for Atelectasis Detection in X-ray Images

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GROUP 25

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1 Introduction

Atelectasis is a respiratory condition that refers to the (partial) collapse of a lung. The condition occurs when the *alveoli* (tiny air sacs in the lung) lose air [22]. A blocked airway can cause *obstructive atelectasis. Non-obstructive atelectasis* can be caused by pressure from outside of the lung.

Atelectasis is extremely common after surgery, with about 90% of patients developing Atelectasis after general anesthesia during surgery [8]. This is because general anesthesia during surgery changes the breathing pattern and exchange of air gases. There are multiple causes for Atelectasis. However, the common cause of Atelectasis after surgery is due to the creation of a mucus plug. This happens because a patient cannot cough or swallow during general anesthesia. Additionally, the medications used to induce anesthesia cause patients to breathe less deeply, which allows mucus to accumulate in the airways, where it would normally be expelled from the lungs [22].

If Atelectasis caused by a buildup of mucus goes unnoticed, it could potentially lead to *Pneumonia*, which occurs when you have an infection that inflames the alveoli. Therefore, atelectasis must be discovered and treated, especially when considering that patients are usually quite vulnerable and weak after major surgery. If a patient is recovering from major surgery, an infection in their airways could majorly worsen their condition. Usually, the diagnosis is made by a specialist based on X-ray images of the lungs. However, an automatic diagnosis system would speed up this process. Algorithmically speaking, some valuable research has already been done about using PSO to detect Pneumonia (see section 3 'Related Work').

We would like to extend the current research by assessing whether we can use PSO for lung x-ray images to detect (partial) lung collapse (Atelectasis) *before* the alveoli become infected, resulting in Pneumonia. We are particularly interested in how early after surgery the Atelectasis can be detected, to prevent any further damage to the lungs. To the best of our knowledge, PSO has not yet been applied to X-ray images for detecting Atelectasis. With this research, we hope to extend upon the current knowledge base to further mitigate Pneumonia caused by the buildup of mucus in the airways after surgery.

This research is also interesting and valuable from a scientific point of view. Firstly, it will asses whether PSO combined with a statistical distinguishing function can be early applied to detect the Atelectasis condition before further complications. This demonstrates the capabilities of particle swarm optimization techniques and proves their usefulness in medical image segmentation. Secondly, if the distinguishing function outperforms or has a similar performance to the CNN, it will potentially prove that there is no necessity for intelligent solutions based on machine learning models for this medical task, while a rule-based statistical method provides more explainability and handles medical data limitations.

2 Background

2.1 Medical image classification: methods and limitations

Nowadays, the image classification and segmentation tasks are almost all handled by machine learning (ML) methods, which mostly include convolutional neural networks (CNNs) architectures [6]. However, these methods face a series of limitations essential to address when the use case is in the medical field and despite promising results, rarely achieved a real-life clinical impact [24]. This is due to several issues: dataset quality (noise or images produced with different machines), size (medical datasets are usually small and very unbalanced) and accessibility (due to privacy regulations, most of them are not public), model interpretability (ML is like a black-box, but medical diagnosis require proof and explanations), lack of generalisation (networks parameters are highly dependent on the dataset) and bias [25]. Hence, a rule-based or statistical method could overcome these limitations, being a more trustworthy approach in healthcare.

2.2 Introduction to Particle Swarm Optimization

The Particle Swarm Optimization algorithm (PSO) [4][10], is a population-based algorithm inspired by the behavior of 'swarms', like birds and schools of fish. PSO is particularly effective in solving continuous optimization problems by changing the positions of particles based on their own experiences and those of their neighbors. However, PSO can be computationally demanding, especially in high-dimensional problems or it could get stuck in local optima.

In this research, we use PSO clustering to attain image quantization in medical images, specifically x-ray images of lungs. The idea is that by performing quantization on the grayscale colors in an x-ray image, we can distinguish abnormalities by performing calculations on the color division. Whilst doing this, we want to keep visual integrity, that is, the image needs to be as distinctively recognizable as the original as possible.

PSO can be effectively used in this sense as it, intuitively, involves grouping similar colors and representing them with a reduced set of representative colors, which, in its formulation, is just a definitional transformation into a clustering problem.

2.3 Formal definition of PSO

At each iteration t, the velocity and position of each particle i are updated as follows:

$$v_i^{(t+1)} = \omega v_i^{(t)} + c_1 r_1 (\hat{x}_i - x_i^{(t)}) + c_2 r_2 (\hat{g} - x_i^{(t)}) \tag{1}$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)} \tag{2}$$

where:

- $x_i^{(t)}$ is the position of particle i at iteration t.
- $v_i^{(t)}$ is the velocity of particle i at iteration t.
- $\hat{x}_i$ is the best position found by particle i so far (personal best).
- $\hat{g}$ is the best position found by any particle in the swarm (global best).
- ω is the inertia weight, controlling exploration vs. exploitation.
- c_1, c_2 are acceleration coefficients for cognitive and social learning.
- r_1, r_2 are random numbers drawn from a uniform distribution in $[0, 1]$.

2.4 Area under ROC curve (AUC)

The ROC curve exhibits the True Positive Rate (TPR) against the False Positive Rate (FPR)[1]. In the context of detecting disease in patients, this would mean the number of actual sick patients labeled as such, against the number of healthy patients labeled as sick. Alternatively, one could use a *confusion matrix*, in which also the True Negatives and False Negatives are displayed.

3 Related Work

3.1 PSO for medical image segmentation

Abdallah et. al. proposed combining Particle Swarm Optimization (PSO) with histogram equalization for segmenting lung CT scans and chest X-ray images. The research claims that using PSO for image segmentation enhances the segmentation accuracy, especially when dealing with complex lung images [1]. Furthermore, Lakshman Narayana et. al. [23] combined PSO with feature extraction methods to perform segmentation on chest x-ray images.

Feng et. al. performed research on using PSO to do image segmentation on infrared images with a low signal-to-noise ratio (SNR). This is relevant to our research, as the x-ray images may also be of low SNR values.[2]

Saifullah et. al. apply PSO combined with a histogram equalization preprocessing to CT scans and X-rays, showing that the PSO approach has a competitive performance with Otsu's and K-means methods, especially on X-ray images [20]. This finding is strengthened by Yu's work [28], who applied PSO to image segmentation for liver fat detection and got better results than using Otsu, Watershed algorithm or K-means clustering.

Saeidifar et. al. used and tested six evolutionary algorithms (PSO, artificial bee colony, genetic clustering, differential evolution, harmony search and gray wolf optimization) to do brain MRI image segmentation to identify tumors. They also compared these methods with K-means and Otsu. Results showed that PSO had the best performance compared to the other methods [19].

3.2 PSO optimization techniques

Kaur et. al. [9] provides an overview of optimization techniques for PSO. They discuss optimization techniques with promising results, of which we have selected a few seemingly interesting ones to draw inspiration and potentially optimize our PSO implementation. We highlight them below.

Thresholding approaches. Yin et. al. [26] used cross entropy for obtaining a threshold for PSO. This reduces the execution time of the image segmentation algorithm.

Nakib et. al. [16] used PSO as an optimization for two-dimensional survival exponential entropy to solve an image segmentation problem in magnetic resonance imaging (MRI) images.

Yu et. al. [27] uses an image segmentation method based on maximum entropy thresholding and quantum behaviour of PSO. They mention that this improved method helps deal with early convergence issues and problems related to large calculation of multi-thresholding segmentation.

Fuzzy System approaches. Masooleh et. al. [15] researched a combination of a fuzzy system with PSO for image segmentation. They apply a set of fuzzy rules to assign each pixel to a colour class. Specifically, they use a 'Hue, Saturation, and Lightness' (HSL) system for pixel classification. In order to

[1] https://en.wikipedia.org/wiki/Receiver_operating_characteristic, accessed at 2025-05-30.

[2] https://www.sciencedirect.com/science/article/pii/S0167865504003289?casa_token=f08ovnGTjVIAAAAA:MVtmkDz-p59frnfqc7CX_BWB3hEdRx4rZOYibFjdxqdKrrW2_QRo6oYzVqCnzeYO-gAEdkMpJQ

reduce a large number of fuzzy rules, PSO is used to automatically produce the least number of fuzzy rules that still provide good classification.

Gopal et. al. [7] used two-phase MRI segmentation. In the first phase, they pre-processed and enhanced the MRI images, removing labels and x-ray marks. Next to that, a median filter was utilized for removing high-frequency components. Finally, a fuzzy c-means (FCM) method is used to calculate an adaptive threshold, which is then optimized with PSO.

Genetic algorithm approaches. Forouzanfer et. al. [5] proposed a combination method of genetic algorithms with FCM clustering that was deemed promising for MRI images. Specifically, they use a 'breeding swarm' algorithm that combines PSO with a genetic algorithm. It was implemented in such a way that PSO facilitates local search, whilst the genetic algorithm is responsible for global search.

Kole et. al. [11] also proposes using a combination of genetic algorithms(GA) with PSO. In this case, PSO-based dynamic clustering is used to find the optimal number of clusters and GA to improve the PSO best particles.

Wavelet approaches. De et. al. [3] integrated PSO into wavelet mutation to prevent getting stuck in local optima. They use a histogram of MRI images and apply entropy maximization to obtain a likely threshold grey level range for segmenting MRI scans.

Clustering approaches. Omran et. al. [17] used binary PSO clustering to properly optimize the number of clusters for k-means clustering.

Liu et. al. [14] proposed using PSO-based fuzzy clustering to sonar image segmentation, in which the images are of low SNR values. They mention that their method has a high convergence speed. Additionally, they use a fuzzy measure and fuzzy integral in their fitness computation.

PSO embedded in neural network approaches. Lian et. al. [13] segmented MRI images based on modified adaptive probabilistic Neural Networks (MAPNN), which incorporates a self-organizing map (SOM), modified particle swarm optimization (MPSO), and a probabilistic neural network ('PNN', used for classification) for segmentation. The MPSO is used to as the smoothing factor σ (i.e., the bandwidth of the Gaussian kernel) for the PNN. This means that the MPSO controls how much gets generalized instead of focusing on individual data points.

PSO can be used to extract features from CNN models. Ladani et. al. used a XOR-based PSO for RegNet feature selection in detecting Pneumonia [12], achieving a 98% accuracy. Pramanik et. al. used PSO feature selection for the same task, but with a ResNet50 model [18]. Additionally, PSO can be used for model's hyperparameter tuning as Aguerchi et. al. used in mammography breast cancer detection [2].

3.3 Why choose PSO in our case?

The above literature review shows promising results in using PSO for medical data segmentation. Moreover, they claimed better results by applying PSO than other methods based on evolutionary algorithms or common clustering methods as K-means. The literature also shows optimisation methods for PSO to overcome the local optima stuck and computational time issues. Thus, we believe it to be a promising method in our case. Moreover, we identified an additional strength of PSO: the objective function can be customized. Based on the problem, we can adapt this function to achieve a particular segmentation. We are not particularly interested in embedding PSO into CNNs as we discussed their limitations. However, we are interested in seeing if pre-processing the images with PSO would improve the CNN as PSO improved the CNN by feature extraction or parameter optimization.

4 Methods

Our methods mainly consist of creating a pipeline for Atelectasis detection. We create a few datasets to serve our scope based on a larger lung X-ray dataset. PSO is used for lung x-ray segmentation and the output is fed to a differential function that uses statistical methods will identify Atelectasis based on lung symmetry, ratio and pulmonary capacity. To determine whether this method is promising compared to CNN-based methods (which have plenty of issues as described above), we also evaluated a CNN model in two scenarios: we gave as input the raw x-ray images or the PSO pre-processed ones. Our hypotheses are: if the rule-based method has a not that different performance as the CNN ones, given the CNN limitations, our method would be more reliable and not need a machine learning approach. Additionally, if CNN trained on PSO images performs better, PSO may help learning in small models by already pre-processing the images and help generalization by limiting the number of shades in the image.

The full implementation of our methods can be found on our Github repository: https://github.com/WizeRacoon/naco-project

4.1 Inspecting the data

The x-ray images used in this project were obtained from the 'NIH Chest X-rays' dataset from Kaggle[3]. The dataset contains 112,120 X-ray images with disease labels from 30,805 patients, and was made by the National Institutes of Health (NIH) of the United States of America[4]. In the description of the dataset, it is mentioned that the disease labels are expected to be over 90% accurate. To create the disease labels, the authors used Natural Language Processing to perform

[3]https://www.kaggle.com/datasets/nih-chest-xrays/data/data, accessed on 2025-04-24.

[4]https://www.nih.gov/, accessed 2025-04-25.

text analysis on disease classifications from the associated radiological reports.

For intermediate testing, we downloaded a sample of the dataset which contained a total of 5606 images, of which 508 were of Atelectasis. So approximately 10% of the total images were of our target disease, Atelectasis. Moreover, 62 images of the sample dataset were of Pneumonia (advanced stage of Atelectasis / lung infection). 3044 of the sample images were of class 'no findings', likely meaning that those were of healthy patients.

The dataset is a collection of unique X-ray images of actual patients. However, not all images were taken from the same angle. When scrolling through the dataset, we noticed some of the images were more zoomed out than others, and we even found images that were slightly tilted and/or had text on the outer side of an image.

When working with the aforementioned Kaggle dataset, we have to keep the following in mind:

- *AP versus PA position:* In x-ray images, there is a difference in viewing position PA and AP (image shot back-to-front of front-to-back). Posterior-anterior (PA) is the standard position, and Anterior-posterior (AP) images are used when a patient cannot make a standard x-ray image, for example, if they cannot stand up by themselves. AP images are said to be of lesser quality, and the heart appears bigger as well[5][6]. Therefore, it would be better to focus on X-ray images taken from a PA position.
- *Zoomed-out or -in images:* We are going to measure the capacity to the lung in relation to the capacity of the body exposed. Differences in lung-size could be due to the way that the image has been shot, but lung-size also just differs per person. Therefore, measuring the lungs as a percentile of the total body exposed in the image will give us the best indication of lung abnormalities.
- *Tilted images:* Since we will be measuring the lung size in relation to the body size on the image, it does not matter for the distinguishing function if the image is tilted. (However, it might matter to the CNN.)
- *Images with text:* Some of the images in the dataset contain text or arrows. It is preferable to filter these out. For the aforementioned sample dataset of 508 images, it might be doable to do this by hand, but for the full dataset, this is infeasible. For now, we consider this mitigating this out of scope for the project, as we would have to train some sort of neural network to spot and remove these. However, we still need to consider this when reflecting on the results.
- *Images of the same patient at different times:* Most of the patients are unique. However, we found some instances of multiple X-ray images belonging to the same patient. The time at which the X-rays were taken of the same patient is different. For example, we found two X-ray images belonging to the same patient, in the first one the patient was 72 years old, and in the second one the patient was 82 years old. The 82 years old image appears earlier in the dataset, but this is likely due to the order in which the radiological reports were scanned by the Natural Language Processor.
- *Multiple diseases for one patient:* Often in the dataset, if a patient has Atelectasis, then they also have at least one other disease. This makes for the following question: *'How can we be sure that our methods are applied for Atelectasis instead of another disease?'*

4.2 Preprocessing the data

We are interested in images that are labeled with 'Atelectasis' and 'No Findings'. Furthermore, by eyeballing the dataset, we have found that indeed the images taken from the PA position seem to be clearer as opposed to the other. This observation is supported by existing information and literature as well. In order to properly evaluate the data, we have created the following datasets:

- **Dataset A:** Contains only images with a PA position. Diseases classified as 'No Findings' and 'Atelectasis and others'.
- **Dataset B:** Contains images with both PA and AP position. Diseases classified as 'No Findings' and 'Atelectasis and others'.
- **Dataset C:** Contains only images with a PA position. Diseases classified as 'No Findings' and 'Only Atelectasis'.

The exact dataset information can be found in the Appendix in table 5. It can be seen that the division of labels is close to 50-50, as to not skew the classification methods. Links to the datasets on Kaggle can be found in Appendix D.

In order to create the datasets, we take the following files from the Kaggle dataset as input: the full dataset (as given on Kaggle), a CSV file with all the entries and their given labels, and a list of train, validation, and test entries as text files. We select the entries that are in the text files, and filter on images that have labels 'No Findings', 'Atelectasis' (pure Atelectasis or also other diseases), 'PA' position, and 'AP' position. We shuffle the intermediate datasets using the Python 'random' library and a random seed of 42.

We also implemented functionality for balancing the datasets such that the CNN has no reason to favor one classification label over the other. The full scripts used for dividing the datasets can be found on our Github repository[7].

[5]https://www.radiologymasterclass.co.uk/gallery/chest/quality/chest-x-ray--ap,accessedat2025-05-30.

[6]https://introductiontoradiology.net/courses/rad/cxr/technique3chest.html, accessed at 2025-05-30.

[7]https://github.com/WizeRacoon/naco-project

4.3 PSO based image segmentation

We use PSO as an image segmentation tool for X-ray lung images. X-ray images are gray-scale images and PSO will segment each image to reduce the number of colors (gray shades). The aim of using PSO in this project is to facilitate the identification of lungs shape and size from X-rays. Usually, on quality X-rays, healthy lungs appear as darker regions surrounded by lighter regions. Thus, applying PSO segmentation to extract a 3-color gray-scale palette should produce 3 clusters as follows:

- Darker shade: lungs and image background
- Intermediate shade: the lung edges, the tissue immediately surrounding the lungs, lung cysts
- Lightest shade: the heart, bones, other tissue (e.g.muscles)

Thus, the lungs could be easily identified as being darker regions surrounded by lighter ones.

In this scenario, the PSO algorithm works as follows:

- Randomly initialize a swarm of N particles with each particle being a KxN array of values in range [0,255] where K is the number of clusters (gray-shade colors) and N the number of colors in each cluster.
- Randomly initialize positions and velocities for each particle
- A local and global best are kept to store the best solution at each iteration, respectively, over all iterations.
- The position and velocity of each particle are updated at each iteration using the formula:

$$v_i^{(t+1)} = \omega v_i^{(t)} + c_1 r_1 (\hat{x}_i - x_i^{(t)}) + c_2 r_2 (\hat{g} - x_i^{(t)}) \quad (3)$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)} \quad (4)$$

  where:
  - $x_i^{(t)}$ is the position of particle i at iteration t.
  - $v_i^{(t)}$ is the velocity of particle i at iteration t.
  - $\hat{x}_i$ is the best position found by particle i so far (personal best).
  - $\hat{g}$ is the best position found by any particle in the swarm (global best).
  - ω is the inertia weight, controlling exploration vs. exploitation.
  - c_1, c_2 are acceleration coefficients for cognitive and social learning.
  - r_1, r_2 are random numbers drawn from a uniform distribution in $[0, 1]$.
- The image pixels are assigned to the closest cluster based on the minimum squared Euclidean distance between the pixel and cluster values
- The fitness of each particle is assessed by summing the above-mentioned Euclidean distances. A lower value means a good fitness in this case
- Update the global best if the case and repeat the above steps until the stopping criteria are met.

Parameter/Criteria	Explanation	Value
N	Number of colors in each cluster	3
K	Number of clusters	3
n	Number of particles	20
c_1, c_2	Acceleration coefficients	1.5
ω	Inertia weight	0.73
Iterations	Number of iterations	5, 20
Stopping criteria	Max. iterations	5, 20
Bounding strategy position	Clip	[0,255]
Bounding strategy velocity	Clip	[-50,50]

Table 1. Criteria, parameters and their values for PSO implementation

Bounding strategy and stopping criteria Working with pixel values, we have to ensure that all the computation results will lay in the [0, 255] range. In one of our previous assignments, we tested various bounding strategies and based on those results, a clipping approach would be appropriate in this case because it provided the best results. Clipping means transforming each out-of-range value to its closest boundary in the specified range. For positions, the range is [0,255], while for velocities it is [-50,50] (allowing particles to turn back if a high value is not good).

For the stopping criteria, we choose to stop PSO when it reaches the maximum number of iterations. From preliminary visual evaluation, a few iterations are enough to identify the lungs. This observation made us think that it will likely work in general. However, to assess this hypothesis, whether a small number of iterations is enough, we evaluated the statistical method on PSO images resulting from 5 iterations versus images resulting from 20 iterations.

4.4 Comparison of PSO iterations and distance metrics

In order to figure out whether performing many PSO iterations is possibly beneficial, we will be comparing 10 randomly drawn toy examples when they went through 5 iterations of PSO against when they went through 20 iterations of PSO. We will be going over each pixel in the image, and compare whether this pixel is of the darker shade, intermediate shade, or lightest shade. We define how equal 5 versus 20 iterations of the same image are by calculating a percentage of the number of pixels that are of a similar shade, and dividing this by the total number of pixels within the image.

Furthermore, in order to observe how using a different distance metric changes our outcomes, we will be performing a few tests in which we have replaced the Euclidean distance with the Manhattan distance. Whenever we are using a different distance metric, it will be clearly stated. Otherwise, the reader may assume that we are using the Euclidean distance.

4.5 Genetic clustering on PSO results

We tried to optimize the quality of PSO results with a method inspired by Kole et. al. [11] (see related work). Kole uses PSO to determine the number of clusters and GA to refine the PSO results. We have a fixed number of clusters and we apply a similar GA approach to potentially improve the results.

Our GA based clustering works as follows:
(1) The initial population is composed of all unique global best palettes provided by PSO
(2) we do an elitist selection based on the fitness score, keeping only half of the initial population
(3) the offspring are computed by randomly selecting two parents, doing a single point cross-over, followed by a mutation with a 0.1 rate, which shifts the clusters' centers. The shift is done by multiplying the centers with a negative value in the range [-0.1,0.1] and clipping the value in the range [0,255] if required.
(4) the fitness function is given by Turi's validity index as used in Kole's approach. It measures the clustering quality by comparing intra-cluster compactness with inter-cluster separation using the formula:

$$\text{Validity\_Index} = y \cdot \frac{\text{intra}}{\text{intra}} \tag{5}$$

where *intra* is the average of distances between pixels' value and the value from their assigned cluster and *inter* is the minimum distance between clusters.

The y measures clusters' separation and is given by $y = c \cdot N(2,1) + 1$ where c is an adjustable parameter and $N(2,1)$ is a Gaussian distribution with mean=2 and std=1. We used a c value of 0.1 as in [11]. The index aims to create compact clusters that are far from each other.

To evaluate this optimization, we compare 10 randomly drawn samples that were processed with (1) 5 iterations of PSO and (2) 10 iterations of PSO followed by GA clustering.

4.6 Classification methods

4.6.1 Classification using a differential function. The initial idea for the classification of Atelectasis (and potential other labels for that matter) revolved around the ability to derive a differential function that would be able to calculate the differences between the right and left lung and classify correctly to the associated label. The process would (a.) segment the lungs in the image even further after PSO had been applied, (b.) extract useful values from this (like symmetry and lung capacity), and then (c.) derive and optimize some function which can use these values for classification.

a. Given is the grey-scale PSO image with a certain number of shades of grey (this is conventionally N=3 in our experiment), from which we want to extract only the lungs. Many methods have been tried like Otsu thresholding [21] and applying knowledge about the approximate location of the lungs, however, the data appeared to be too inconsistent for these general methods to work. After a significant amount of time spent to see what works and what doesn't, the most promising (visually seemingly most accurate output images across the board) order of operations found is the following (see appendix C for visual clarification):

1. Crop off the bottom 90 pixels of the incoming PSO image to discard the visual bar with resulting colors, leaving only the original image, with PSO overlay.
2. Shade thresholding is used to separate lighter areas from darker areas and polarize those toward full white and full black. The hope is that the lungs and image background will both be detected as dark and the rest as light. The black and white immediately get inverted as well for the next step. However, as there are multiple shades of grey, it is, across all data, inconsistent which shade are the lungs. This is why the polarization first happens on the darkest shade after which we invert and detect if (1.) it has a certain amount of white (to circumvent picking a too harsh shade as the threshold, likely not picking up the lungs) and (2.) a verticle line exists around the middle of the image that is fully black (the middle between two lungs). If the conditions aren't met, the process is repeated with the threshold being one shade lighter (as shown in Figure 7a), and so on, ensuring that the most optimal shade is selected as the polarization threshold, which definitely includes the lungs.
3. The sides of the image flood with black, such that the background disappears, leaving only the contours of the torso, with the lungs and some noise remaining. Sometimes, the lungs are exposed to the background, making the lungs disappear while flooding. This is why the flooding is capped at 15% of the image size from all borders (see figure 7b).
4. In some images, the ribcage is quite dominant, by expanding each white pixel toward its neighbors, connections can be formed 'across ribs', leaving larger inter-connected pieces (see figure 7c) crucial for step 6.
5. Additional preprocessing before step 6 is drawing a black line in the middle of the image to still create two large segments if the lungs are somehow still attached (see figures 7d). If the lungs are already detached this step is trivial as the line will simply go over the sternum (which is already black).
6. Now the two biggest white contours get calculated, which, by now, should be the lungs, and the rest, which should be residual noise gets eliminated

7. The final segmentation result gets drawn for completion purposes.
8. Calculate a symmetry line which is the average x-position of the centers of the two lungs.

b. The two values extracted which were both feasible from Step 8 of (a) and seemingly useful are:
1. The symmetry percentage, that is, when the image is mirrored on the symmetry line, what percentage of overlap do the lungs have? This could be affected by the input images being slightly rotated. It would be logical to expect a positive correlation between symmetry and healthy lungs versus those with atelectasis.
2. The same train of thought for symmetry percentage applies to the second extracted variable; proportional lung capacity. This signifies (with a multiplier) how much bigger the biggest lung is than the smaller lung.

c. With a balanced training dataset (to avoid majority class bias), the function underneath gets optimized to find the best combination of weights and threshold by maximizing label accuracy:

$$w\_sp * sp + w\_plc * plc = threshold \tag{6}$$

where:
- w_sp is the weight for symmetry percentage.
- sp is the symmetry percentage.
- w_plc is the weight for proportional lung capacity.
- plc is the proportional lung capacity.
- $threshold$ is some threshold.

After, the optimized function gets tested with a test dataset, to prevent overestimation in performance. The training and test sets get randomly picked every time in an 80-20 split.

4.6.2 Classification using a CNN. We found a CNN network on Kaggle that claimed to have a 92.6% accuracy in detecting Pneumonia in lung x-ray images. Since we found in the existing literature that Atelectasis can occur as an early stage of Pneumonia, we first supply our raw X-ray Atelectasis images to CNN to establish a baseline. Afterward, we supply the datasets as specified earlier. A table representation of the CNN architecture can be found in the appendix in figure 5.

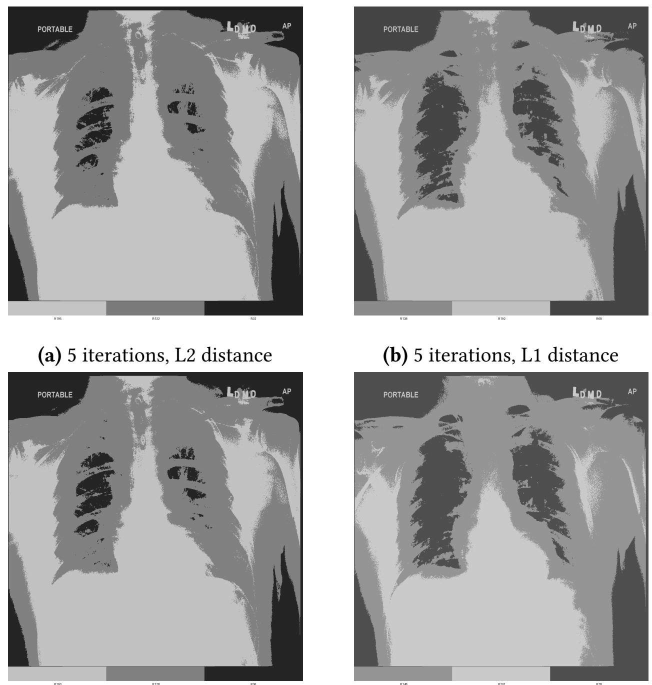
5 Results

5.1 Comparison of PSO iterations and distance metrics

As can be seen in the results, most of the images agree on pixel shade for more than 90% of the time. However, there are some images for which there is a bigger difference noted. For PSO 20 iterations with L2 distance, the biggest difference in pixel shade is in image ‘00001736_002.png’. We can see in figure 2 that the shape of the lungs is indeed quite different

Image index	PSO 5 iter. (L2)	PSO 20 iter. (L2)	GA+PSO 10 iter. (L2)	PSO 5 iter. (L1)	PSO 20 iter. (L1)
00019508_002.png	baseline	96.80%	90.89%	85.23%	73.11%
00016142_036.png	baseline	83.98%	84.44%	89.14%	91.52%
00001736_002.png	baseline	73.77%	91.44%	100.00%	96.23%
00010729_001.png	baseline	97.31%	93.08%	90.34%	91.13%
00017388_000.png	baseline	85.31%	90.56%	88.72%	75.09%
00013177_000.png	baseline	94.20%	94.08%	94.88%	97.18%
00001211_000.png	baseline	91.76%	93.30%	74.70%	90.12%
00006142_002.png	baseline	93.03%	56.02%	82.28%	76.73%
00014253_010.png	baseline	95.91%	94.85%	85.60%	94.59%
00013294_001.png	baseline	94.02%	91.56%	61.38%	68.07%
Average	baseline	90.609%	88.022%	85.227%	85.377%

Table 2. Amount of pixels of a similar shade in percentage between various optimization strategies. The 5 iterations PSO using L2 is considered the baseline.



(a) 5 iterations, L2 distance **(b)** 5 iterations, L1 distance

(c) 20 iterations, L2 distance **(d)** 20 iterations, L1 distance

Figure 1. Image `00019508_002.png` after 2 and 20 iterations, for L2 (Euclidean) distance, and L1 (Manhattan) distance.

after 20 iterations when compared to just 5 iterations. We also see a significant drop in similarity when looking at image ‘fig:00006142_002_.png’, where the similarity rate for 20 iterations is quite high, but the similarity rate when using a genetic algorithm is rather low. Investigating the images in figure 3 shows why; the separate ribs are more clearly defined when there is no use of genetic algorithms. The genetic algorithm smoothes out the region.

The Manhattan distance takes the absolute distance. In figure 1, we can see that using the Manhattan distance (L1) rather than the Euclidean distance (L2) also seems to be a bit less precise and smoother.

In general, it seems like more iterations lead to slightly better results. And, using a genetic algorithm or the absolute distance somewhat smoothes out the harsh edges. It needs to be mentioned that the PSO runs are not stochastic, and therefore, results may vary between runs. However, by investigating a set of toy examples, we can carefully estimate the behavior of the program.

5.2 Classification using differential function

5.2.1 Experiments conducted. Experiments (in the fashion of section 4.6.1) have been conducted (result in table 3) with the following in mind:

- The encompassing dataset used is all computed PSO images (up to and including 20 iterations) of the full sample dataset of x-ray images on Kaggle.
- We train and test the differential function on distinction ability between labels 'No Findings' and the disease group, which can be: 1. only Atelectasis (only A.), 2. Atelectasis and potentially other diseases (A. + other) and 3. any disease(s) (any).
- The above experiments have been conducted on filtered datasets (1.) PSO images of iteration 20, (2.) PSO images of iteration 20 and from the 'PA' viewing position, (3.) PSO images of iteration 5.
- The dataset size has been set to 200 (100 randomly picked 'No Findings' and 100 randomly picked disease instances of the filtered dataset). This makes the dataset both immune to majority bias and consistent in size across all experiments[8].
- To achieve statistical consistency, each experiment's outcome is the average (mean) of 100 trials[9] (with standard deviation included).
- To avoid scaling of the weights, and get comparatively relevant results of w_sp and w_plc (and their respective standard deviations), the *threshold* of equation 6 has been fixed at 100.
- Resolution, statically set[10] to 100, is the degree up until which the differential function does optimize (higher is more accurate but takes longer).
- True positives, true negatives, false positives and false negatives also get calculated with their respective standard deviation over 100 trials.

5.2.2 Interpretation of results. The resulting classification accuracies, barely outscoring random guessing with ranges between 49% and 56% across experiments, indicate that the method does not perform well in practice for classifying Atelectasis from X-ray images given PSO inputs, regardless of the number of PSO iterations and viewing position. This suggests that either or both the extracted features (symmetry percentage and proportional lung capacity) do not capture the essence for reliable diagnosis of Atelectasis or the input PSO images were not segmented to the desired extent / in the desired way. Looking at the further segmentation processes the differential function applies on the first ten PSO images and the resulting lung area it projects (see figure 8), questioning the quality of the incoming PSO images seems to be a fair concern as the area for which the differential function predicts is the lungs is quite accurate in relation to the input.

Proportional lung capacity appears to contribute more to the classification than symmetry percentage. However, the high variance in weight values and prediction outcomes suggests limited generalizability. This inconsistency can be caused by unreliable lung segmentation using PSO, variations in X-ray image quality, and the fact that Atelectasis can occur bilaterally, undermining assumptions of asymmetry.

The comparison between different dataset filters (5 iterations, 20 iterations, 20 iterations and PA only) showed only small differences in classification performance (even though the filters that were projected to do better did indeed do so), showing that (virtual) convergence of PSO and slightly better data quality with 'PA' position is not enough.

The most curious result is that the differential function seems to improve in accuracy when trying to detect diseases other than only Atelectasis. Perhaps other diseases have clearer and more consistent visual clues to asymmetry in the lungs; which the differential function tries to distinguish on.

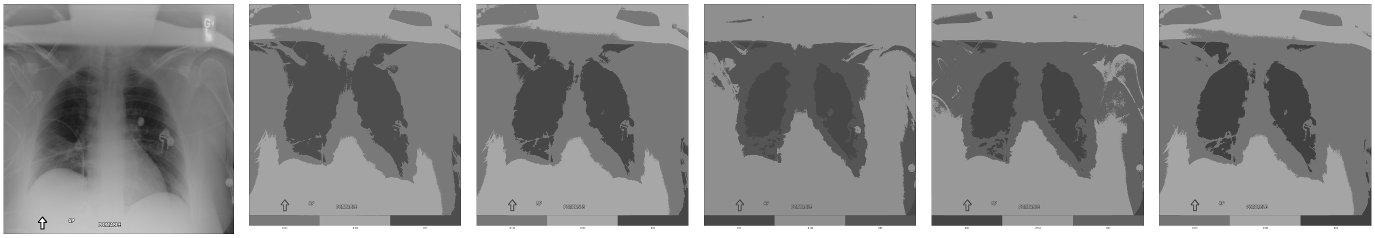
5.3 Classification using a CNN

As can be seen in table 4, the CNN performed worse for datasets with both PA and AP positions (dataset B) as opposed to datasets with only PA positions (dataset A). This is in line with what was found in the literature. Moreover, it can also be seen that the addition of PSO worsens the classification results. Our intuition for this, is that performing color segmentation on the images removes some information that might help the CNN correctly classify the images. More iterations of PSO (20 in this case) seem to be slightly better than just 5 iterations of PSO.

[8] In the sample dataset, there are only 101 images labeled only with disease 'Atelectasis' and have view position 'PA', so that is our bottleneck in uniform dataset size

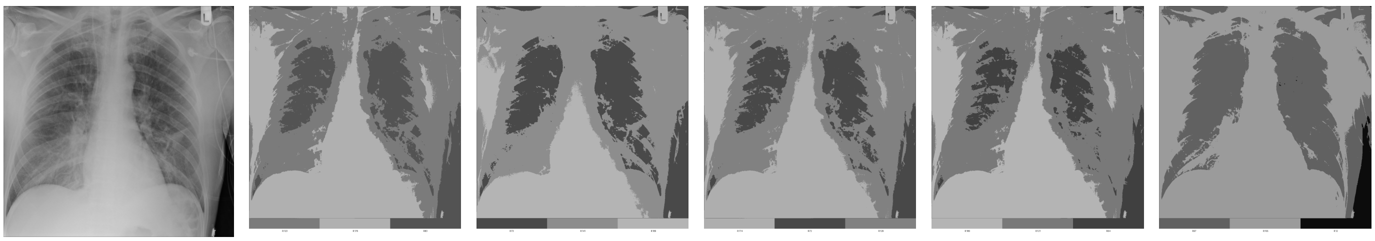
[9] 3 varying experiments with 1000 trials were conducted. These had, within indistinguishable margins, the same results (accuracy, means and standard deviations), as the equal experiment counterparts with only 100 trials, demonstrating that 100 trials are enough for statistical convergence

[10] 3 varying experiments with a resolution of 1000 were conducted. These had, within indistinguishable margins, the same results (accuracy, means, and standard deviations), as the equal experiment counterparts with a resolution of 100, demonstrating that a resolution of 100 is enough for statistical convergence for a dataset size of 200



(a) original image **(b)** 5 iterations **(c)** 10 iterations **(d)** 15 iterations **(e)** 20 iterations **(f)** 10 iterations, GA

Figure 2. Image `00001736_002.png` and its reconstructions after 5, 10, 15, and 20 iterations of PSO, and after 10 iterations of PSO for finding the local optima in a Genetic Algorithm (GA).



(a) original image **(b)** 5 iterations **(c)** 10 iterations **(d)** 15 iterations **(e)** 20 iterations **(f)** 10 iterations, GA

Figure 3. Image `00006142_002.png` and its reconstructions after 5, 10, 15, and 20 iterations of PSO, and after 10 iterations of PSO for finding the local optima in a Genetic Algorithm (GA).

	Iter 20			**Iter 20 (PA only)**			**Iter 5**		
	only A.	A. + other	any	only A.	A. + other	any	only A.	A. + other	any
Train accuracy	57.68%	61.16%	62.30%	57.34%	62.01%	61.39%	57.13%	60.95%	62.29%
Train std	2.280%	2.977%	3.112%	2.392%	2.815%	2.767%	2.392%	3.061%	3.157%
Train # images	160	160	160	160	160	160	160	160	160
Test accuracy	50.38%	55.05%	54.83%	49.00%	55.95%	54.87%	49.00%	54.28%	56.90%
Test std	8.834%	6.937%	6.831%	7.450%	8.061%	6.985%	7.450%	7.502%	8.661%
Test # images	40	40	40	40	40	40	40	40	40
TP	9.090	9.620	9.450	10.46	9.840	9.090	9.370	9.810	9.880
TN	11.06	12.40	12.48	9.140	12.54	12.86	10.66	11.90	12.88
FP	9.230	8.010	7.080	10.96	7.340	7.360	9.020	8.110	7.100
FN	10.62	9.970	10.99	9.440	10.28	10.69	10.95	10.18	10.14
TP std	4.285	4.291	4.486	4.451	4.789	4.157	4.089	4.310	4.304
TN std	4.742	3.882	3.583	3.749	3.628	3.945	4.054	4.075	3.906
FP std	4.578	3.424	3.246	4.155	3.646	3.772	3.859	3.917	3.743
FN std	4.672	4.064	4.295	4.105	4.353	4.330	4.217	4.182	4.336
w_sp **mean**	30.65	29.58	32.90	28.60	30.60	27.56	34.56	27.62	30.01
w_sp **std**	29.51	25.78	23.17	28.00	22.80	25.00	29.82	24.85	24.18
w_plc **mean**	63.30	63.20	60.35	68.17	64.34	65.67	60.61	64.91	61.80
w_plc **std**	19.68	17.43	16.10	21.03	16.19	17.50	20.56	16.66	16.75
Threshold	100	100	100	100	100	100	100	100	100
Dataset size	200	200	200	200	200	200	200	200	200
# Trials	100	100	100	100	100	100	100	100	100
Resolution	100	100	100	100	100	100	100	100	100
PSO Particles (N)	3	3	3	3	3	3	3	3	3
PSO Iterations	20	20	20	20	20	20	5	5	5

Table 3. Differential function results. TP = true positives, TN = true negatives, FP = false positives, FN = false negatives.

Configuration	CNN Accuracy	CNN Loss
A raw	67.05%	0.641
A PSO 5 iterations	60.62%	0.659
A PSO 20 iterations	61.41%	0.683
B raw	61.07%	0.684
B PSO 5 iterations	58.94%	0.690
C raw	57.53%	0.664
C PSO 5 iterations	55.87%	0.673

Table 4. CNN accuracy and loss for different datasets after 30 epochs. Results are averaged over 9 runs, in which for every 3 runs a new test folder was used (test_1, test_2, test_3).

6 Discussion

Unfortunately, we found that for an almost perfectly balanced dataset, the addition of preprocessing using PSO does *not* improve classification for the Atelectasis disease. We can see in table 4 that using PSO images as input does not beat the quality of regular x-ray images. Classification through the use of a differential function may be possible. Although it has become clear to us that defining such a function is quite the challenge in itself, as the colors are sometimes segmented in a way that makes it difficult to do any sort of further processing on the image. A more extensive study would need to be done on improving the differential function. Which is also why we propose it for future work.

Moreover, we believe that the quality of the images highly impacts the ability to perform good classification. For example, in some of the X-ray images, text and symbols were present, or they were slightly tilted. Although ideally, we would want to be able to perform classification on badly taken images as well, it was simply not realistic to achieve such a goal in this project. The PSO step performed well for the good quality x-ray images, mapping the lungs to a shade different than the surrounding tissue and background. Proving that PSO could be used for outlining a lung when the x-ray image is of good enough quality.

Furthermore, since Atelectasis can occur as an early stage of Pneumonia, it could be that our CNN made for detecting Pneumonia cannot pick up the correct cues yet. Since Pneumonia inflames the alveoli, it might simply be more visible on X-rays.

Another possible reason for the failure of our methods might be that Atelectasis can actually occur in both lungs simultaneously[11]. An X-ray image containing a partial lung collapse on both sides might mess up the similarity calculation in the differential functions, or trick the CNN.

Lastly, it is important to think about whether using PSO for image segmentation is even a good idea at all when trying to optimize CNN classification performance. Other works have used PSO, for example, to make improvements in features for neural networks. It could be that segmentation as a form of preprocessing for CNN classification just requires an extensive amount of fine-tuning, which is a difficult thing to do.

[11] https://www.verywellhealth.com/understanding-atelectasis-2248927, accessed at 2025-06-02.

7 Future Work

7.1 Using PSO to segment x-rays for other diseases

Due to time constraints, we have only performed the described methods on Atelectasis. Looking at the results, we were unfortunately not successful to meaningfully capture the essence of the lungs with PSO (see figure 4). However, we mentioned before that this could be because Atelectasis might just occur be too early to be clearly visible on X-rays.

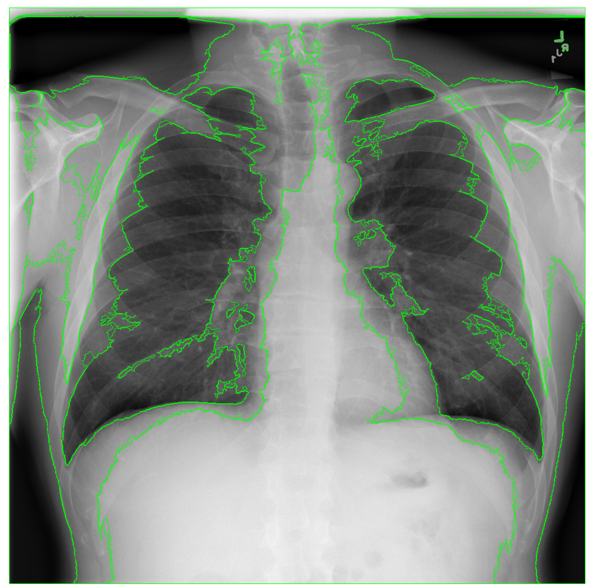


Figure 4. PSO contours over original X-ray, 50 iterations (virtually converged), 00001121_000.png. Clearly, The PSO has trouble defining the subtle edges of the lung, even though it can distinguish the general shape.

7.2 Trying out different optimization techniques

Firstly, the literature shows that much work is done in getting the color segmentation right. If given more time, we could have tried out different methods described in the related work to see if applying an accuracy optimization on our PSO results in better segmented images.

Secondly, there is also potential for other optimizations. For example, the CNN could have been improved upon. We did not dive deeply into the topic of neural networks, because this simply was not the focus of our research.

Thirdly, using a CNN and a distinguishing function are not the only methods for classifying images. There might be other methods for automated classification that are better suited for the task at hand. We propose further research in this area.

Lastly, improving the differential function in such a way that we got distinguishable results was quite a time-consuming task. Although difficult, we did manage to get continuously better. This shows that it might be possible to further tweak this function to get optimal classification results. Further research includes answering questions including, but not limited to:

1. Is symmetry percentage and proportional lung capacity equal in its distinguishing capability, or is one even a subset of another)?
2. With a dataset of perfect PSO input images (where the full lung is exactly differentiated), how good is the accuracy? Could it even surpass a neural network or is the technique inherently capped out at a certain percentage?
3. Are there more meaningful parameters extractable that would improve the differential function?
4. Given the classification using a distinguishing function seemed to perform better on a dataset with both Atelectasis as well as other diseases, is this function perhaps better on other diseases?

7.3 Making the project more lightweight using JPEG

Currently, images in this project, whether from the original dataset or generated by us, are in the '.png' format. This seemed the safe option as this format is lossless, which, might have been important due to the potential accuracy diminishment of a lower-quality image. However, we have not shown this. The lossy compression image format JPEG (or '.jpg') may perform indistinguishably well. This is worthwhile to consider as, due to us working with greyscale images only, JPEG compression can be up to 10x smaller in saving size (tested by converting a few images to '.jpg'), making the whole project considerably more lightweight in disk space.

8 Conclusion

As can be seen by the results of our experiments, PSO seems to be quite hard to adjust for lung x-ray segmentation. It turns out that tweaking the program in such a way that we can make a clean cut-out of the lungs is difficult. There are several factors at play that influence the results, such as patients having more than one disease, or the presence of two partial lung collapses in one image. It should also be taken into account that Atelectasis might just not be as detectable of a disease as Pneumonia, which could explain why there is much research done in Pneumonia detection (see related work section), and to our knowledge none in Atelectasis. Although we do, to some degree, still believe that there is potential, as numerous successful experiments have been performed for Pneumonia, it should be taken into consideration that segmenting the image by color causes a loss of valuable information. This can be problematic when the lost information is needed for a distinguisher to correctly classify the image.

9 Author contributions

Andreea Gabrian did all of the work on implementing the PSO, and Genetic algorithm optimization with PSO.

Mika Sipman made the distinguishing function and performed the distinguish function experiments. Mika Sipman also restructured the codebase in such a way that it can all be executed by use of the atelectasis.py script only.

Lisanne Weidmann created the balanced datasets, executed PSO iterations on iCIS clusters, performed experiments on the CNN and in comparing different PSO iterated images. Cleaned up the Github repository and merged Andreaa's PSO+GA with Mika's restructured PSO+distinguish function codebase.

We were all quite content with the group work, and also helped eachother out when needed.

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A Dataset label information

Split	Class	Dataset A				Dataset B				Dataset C			
		PA	AP	Atelectasis	No Finding	PA	AP	Atelectasis	No Finding	PA	AP	Atelectasis	No Finding
train	NORMAL	3493	0	0	3493	1746	1746	0	3492	1511	0	0	1511
train	ATELECTASIS	3493	0	3493	0	1746	1746	3492	0	1511	0	1511	0
val	NORMAL	874	0	0	874	437	437	0	874	378	0	0	378
val	ATELECTASIS	874	0	874	0	437	437	874	0	378	0	378	0
test_1	NORMAL	453	0	0	453	211	245	0	456	107	0	0	107
test_1	ATELECTASIS	453	0	453	0	217	233	450	0	107	0	107	0
test_2	NORMAL	453	0	0	453	232	223	0	455	107	0	0	107
test_2	ATELECTASIS	453	0	453	0	230	221	451	0	107	0	107	0
test_3	NORMAL	453	0	0	453	237	212	0	449	107	0	0	107
test_3	ATELECTASIS	453	0	453	0	231	226	457	0	107	0	107	0

Table 5. Division of labeled images in Datasets A, B, and C.

B CNN Architecture

Below a table representation of the CNN from Kaggle can found.[12]

Layer (type)	Output Shape	Param #
Conv2D (conv2d_1)	(150, 150, 32)	320
BatchNormalization (batch_normalization_1)	(150, 150, 32)	128
MaxPooling2D (max_pooling2d_1)	(75, 75, 32)	0
Conv2D (conv2d_2)	(75, 75, 64)	18,496
Dropout (dropout_1)	(75, 75, 64)	0
BatchNormalization (batch_normalization_2)	(75, 75, 64)	256
MaxPooling2D (max_pooling2d_2)	(38, 38, 64)	0
Conv2D (conv2d_3)	(38, 38, 64)	36,928
BatchNormalization (batch_normalization_3)	(38, 38, 64)	256
MaxPooling2D (max_pooling2d_3)	(19, 19, 64)	0
Conv2D (conv2d_4)	(19, 19, 128)	73,856
Dropout (dropout_2)	(19, 19, 128)	0
BatchNormalization (batch_normalization_4)	(19, 19, 128)	512
MaxPooling2D (max_pooling2d_4)	(10, 10, 128)	0
Conv2D (conv2d_5)	(10, 10, 256)	295,168
Dropout (dropout_3)	(10, 10, 256)	0
BatchNormalization (batch_normalization_5)	(10, 10, 256)	1,024
MaxPooling2D (max_pooling2d_5)	(5, 5, 256)	0
Flatten (flatten_1)	(6400)	0
Dense (dense_1)	(128)	819,328
Dropout (dropout_4)	(128)	0
Dense (dense_2)	(1)	129
Total Parameters		1,246,401
Trainable Parameters		1,245,313
Non-trainable Parameters		1,088

Figure 5. Updated Convolutional Neural Network Model Architecture Summary

[12] https://www.kaggle.com/code/madz2000/pneumonia-detection-using-cnn-92-6-accuracy, accessed on 2025-05-13.

C Differential Function Images

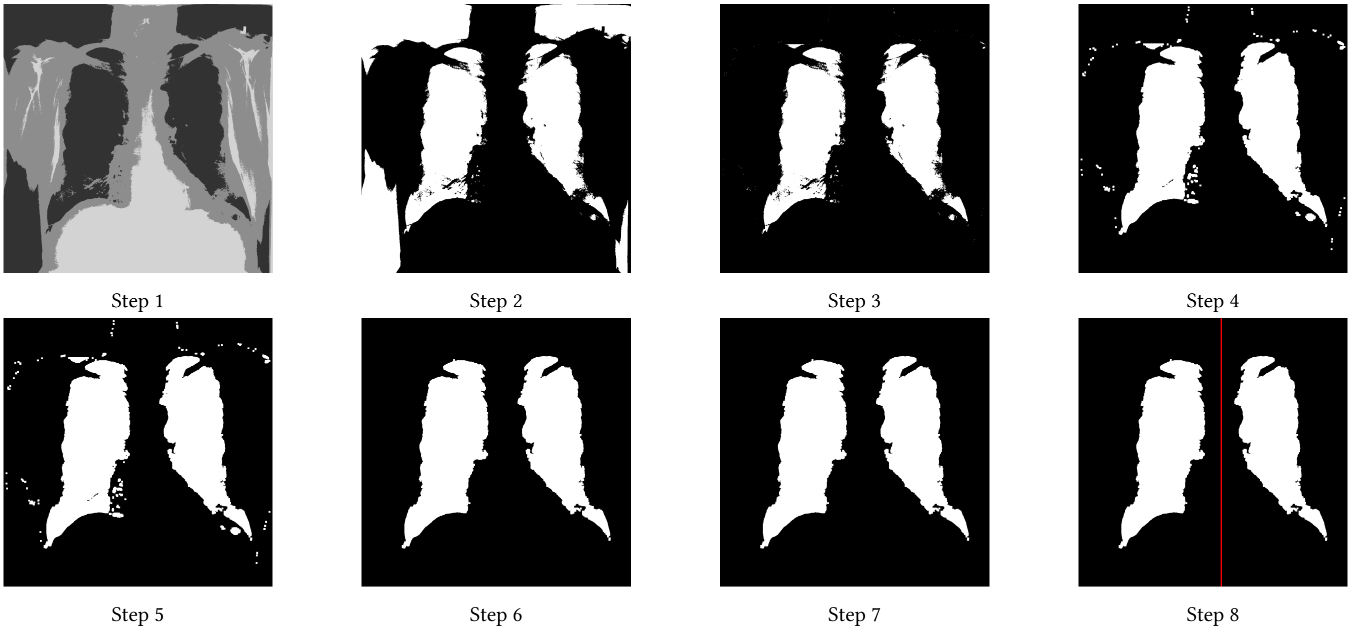
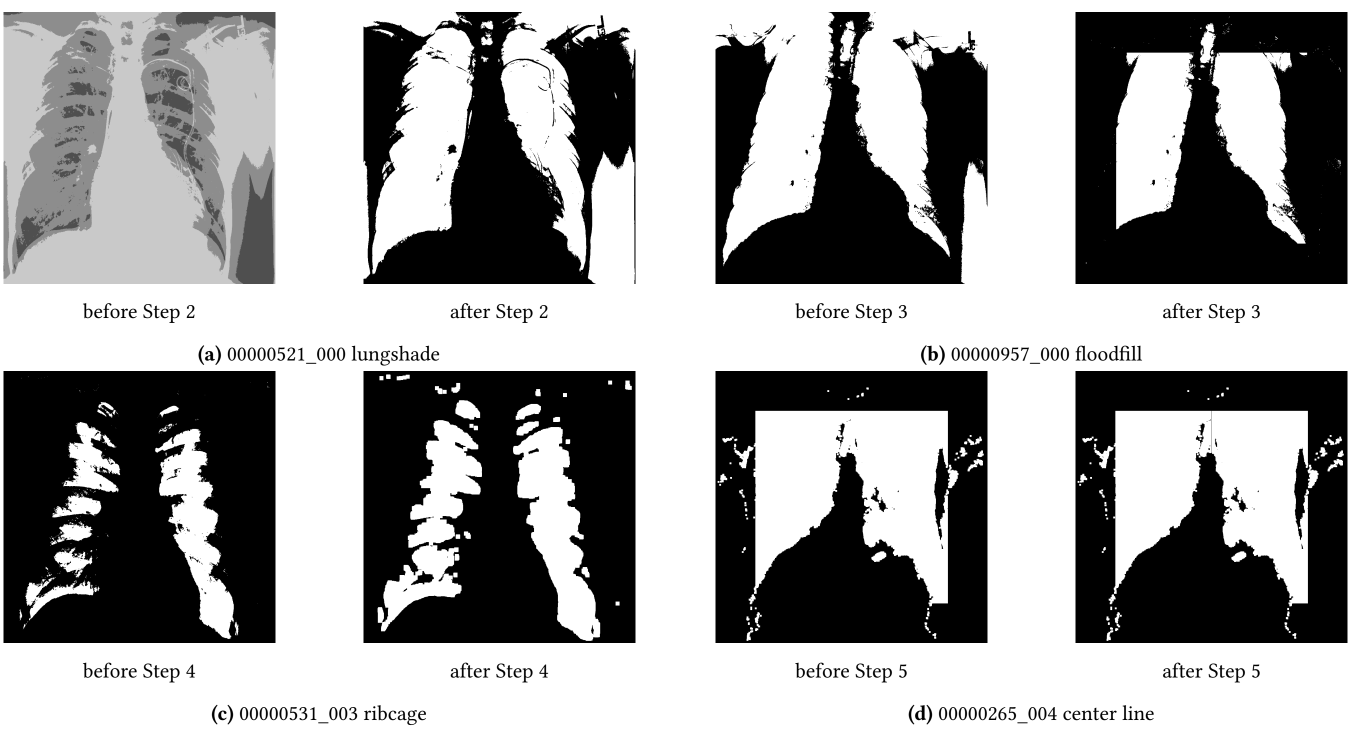


Figure 6. Full transformation of 00000030_001



(a) 00000521_000 lungshade

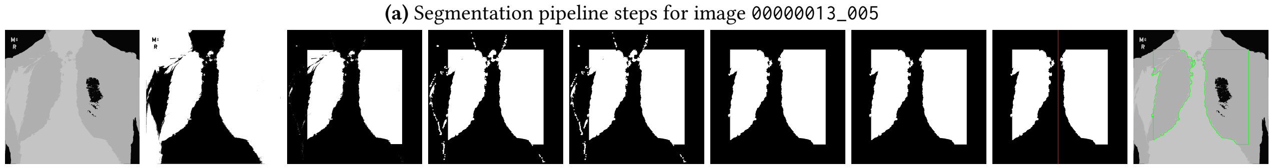
(b) 00000957_000 floodfill

(c) 00000531_003 ribcage

(d) 00000265_004 center line

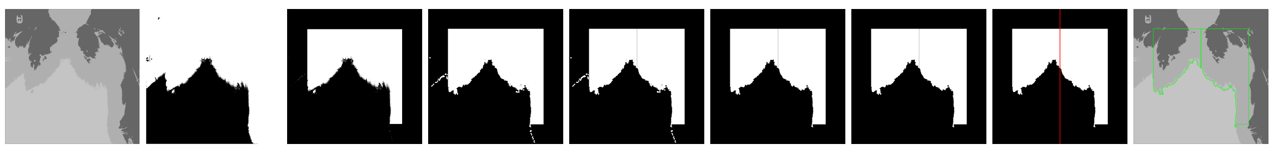
Figure 7. Difficulties

(a) Segmentation pipeline steps for image `00000013_005`

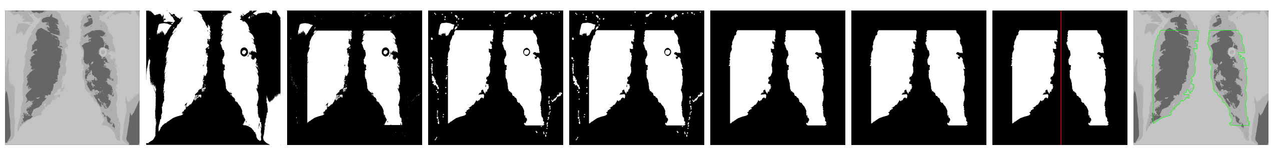


(b) Segmentation pipeline steps for image `00000017_001`

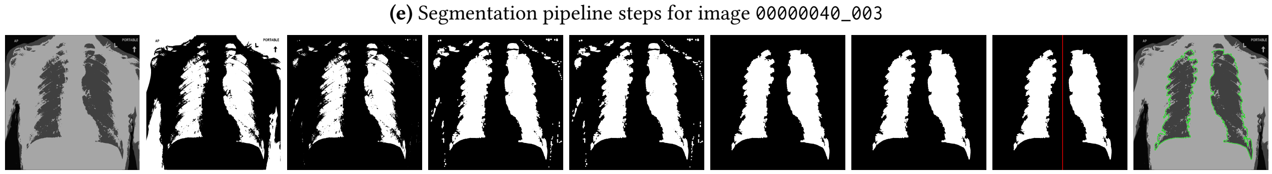
(c) Segmentation pipeline steps for image `00000030_001`



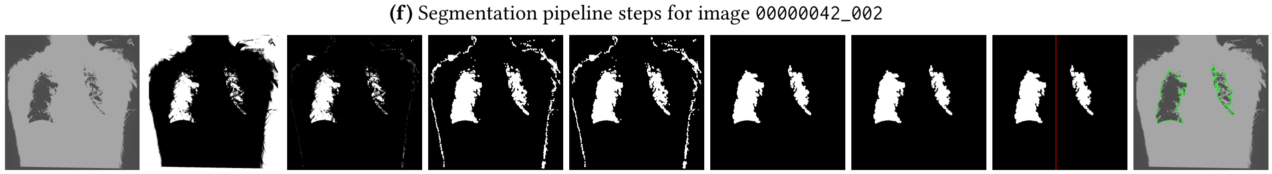
(d) Segmentation pipeline steps for image `00000032_001`



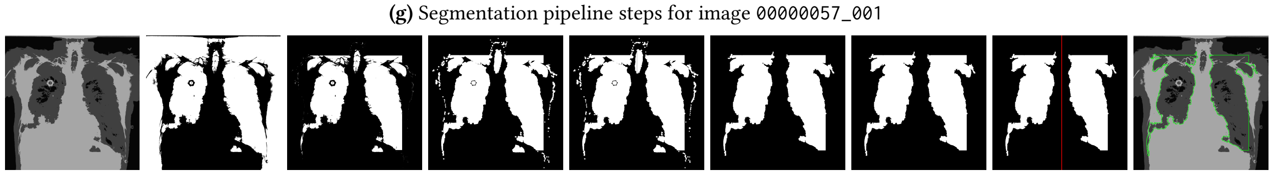
(e) Segmentation pipeline steps for image `00000040_003`



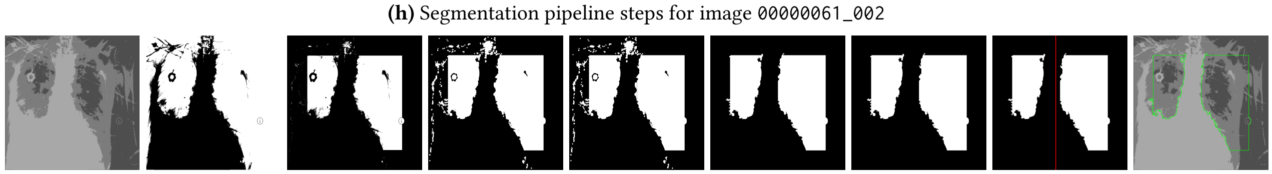
(f) Segmentation pipeline steps for image `00000042_002`



(g) Segmentation pipeline steps for image `00000057_001`



(h) Segmentation pipeline steps for image `00000061_002`



(i) Segmentation pipeline steps for image `00000061_019`

Figure 8. Step-by-step lung segmentation process for first 10 images in sample dataset

D Links to build datasets

Link to sample NIH dataset:
https://www.kaggle.com/datasets/nih-chest-xrays/sample

Link to full NIH dataset:
https://www.kaggle.com/datasets/nih-chest-xrays/data

The PSO iterated image datasets can be found here: https://drive.google.com/drive/folders/1jwt3imVoDML2k_qO91k1oCQ5Nv7b0EbH?usp=sharing
Unfortunately, the raw image datasets were too big to upload to Google Drive (but they can be found on Kaggle, see below).

All balanced datasets (raw and PSO iterated) can be found on Kaggle. Lisanne Weidmann might need to grant you access using your Kaggle username.

Link to dataset A (raw x-ray images):
https://www.kaggle.com/datasets/lisanneweidmann/only-pa

Link to dataset A (PSO 5 iterations reconstructions):
https://www.kaggle.com/datasets/lisanneweidmann/only-pa-pso5

Link to dataset A (PSO 20 iterations reconstructions):
https://www.kaggle.com/datasets/lisanneweidmann/only-pa-pso20

Link to dataset B (raw x-ray images):
https://www.kaggle.com/datasets/lisanneweidmann/pa-ap-atelectasis-normal

Link to dataset B (PSO 5 iterations reconstructions):
https://www.kaggle.com/datasets/lisanneweidmann/pa-ap-atelectasis-normal-pso5

Link to dataset C (raw x-ray images):
https://www.kaggle.com/datasets/lisanneweidmann/only-pa-and-atelectasis

Link to dataset C (PSO 5 iterations reconstructions):
https://www.kaggle.com/datasets/lisanneweidmann/only-pa-atelectasis-pso5